



GCE A LEVEL MARKING SCHEME

**A LEVEL
CHEMISTRY
1410U40 – 1**

About this marking scheme

The purpose of this marking scheme is to provide teachers, learners, and other interested parties, with an understanding of the assessment criteria used to assess this specific assessment.

This marking scheme reflects the criteria by which this assessment was marked in a live series and was finalised following detailed discussion at an examiners' conference. A team of qualified examiners were trained specifically in the application of this marking scheme. The aim of the conference was to ensure that the marking scheme was interpreted and applied in the same way by all examiners. It may not be possible, or appropriate, to capture every variation that a candidate may present in their responses within this marking scheme. However, during the training conference, examiners were guided in using their professional judgement to credit alternative valid responses as instructed by the document, and through reviewing exemplar responses.

Without the benefit of participation in the examiners' conference, teachers, learners and other users, may have different views on certain matters of detail or interpretation. Therefore, it is strongly recommended that this marking scheme is used alongside other guidance, such as published exemplar materials or Guidance for Teaching. This marking scheme is final and will not be changed, unless in the event that a clear error is identified, as it reflects the criteria used to assess candidate responses during the live series.

WJEC GCE A LEVEL CHEMISTRY
PHYSICAL AND ORGANIC CHEMISTRY
GENERAL INSTRUCTIONS

Recording of marks

Examiners must mark in red ink.

One tick must equate to one mark (apart from the questions where a level of response mark scheme is applied).

Question totals should be written in the box at the end of the question.

Question totals should be entered onto the grid on the front cover and these should be added to give the script total for each candidate.

Marking rules

All work should be seen to have been marked.

Marking schemes will indicate when explicit working is deemed to be a necessary part of a correct answer.

Crossed out responses not replaced should be marked.

Credit will be given for correct and relevant alternative responses which are not recorded in the mark scheme.

Extended response question

A level of response mark scheme is used. Before applying the mark scheme please read through the whole answer from start to finish. Firstly, decide which level descriptor matches best with the candidate's response: remember that you should be considering the overall quality of the response. Then decide which mark to award within the level. Award the higher mark in the level if there is a good match with both the content statements and the communication statement. Award the middle mark in the level if most of the content statements are given and the communication statement is partially met. Award the lower mark if only the content statements are matched.

Marking abbreviations

The following may be used in marking schemes or in the marking of scripts to indicate reasons for the marks awarded.

cao = correct answer only

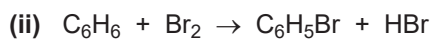
ecf = error carried forward

bod = benefit of doubt

1 (a) (i)

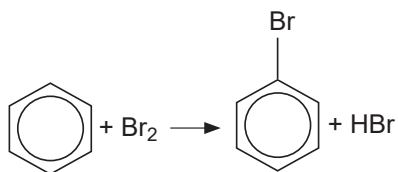
Reaction of benzene	Name of organic product	Name of catalyst
with bromine	bromobenzene	iron/iron(III) bromide/ aluminium bromide [1]
with ethanoyl chloride	phenylethanone [1]	iron(III) chloride/ aluminium chloride [1]

[3]



[1]

or

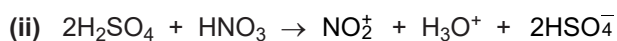


(iii) electrophilic substitution

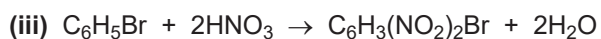
[1]

(b) (i) concentrated sulfuric acid [1]
concentrated nitric acid [1]

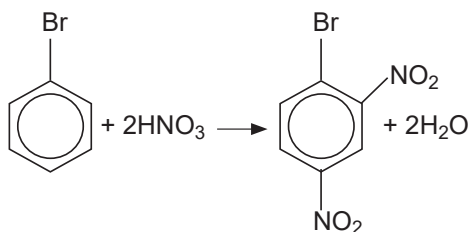
[2]



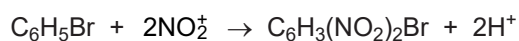
[2]



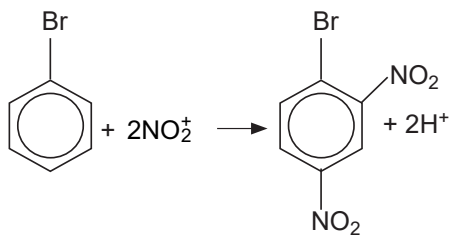
or



or



or



[1]

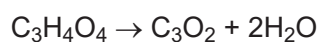
AVAILABLE
MARKS

(a) propanedioic acid

[1]

(b) (i) $\text{HOOCCH}_2\text{COOH} \rightarrow \text{C}_3\text{O}_2 + 2\text{H}_2\text{O}$

or



[1]

(ii) $\text{O}=\text{C}=\text{C}=\text{C}=\text{O}$

[1]

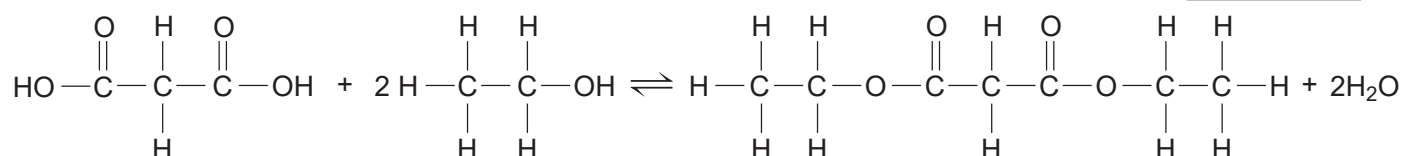
(c) (i) $\text{HOOCCH}_2\text{COOH} + 2\text{PCl}_5 \rightarrow \text{ClOCCH}_2\text{COCl} + 2\text{POCl}_3 + 2\text{HCl}$

[2]

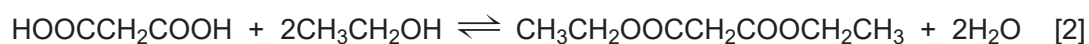
(ii) propanedioyl (di)chloride

[1]

(d) (i)



or



(ii) van der Waals' forces and permanent dipole-dipole attractions
between molecules of ester [1]

stronger hydrogen bonds between molecules of acid [1]

[2]

(e) (i) LiAlH_4

[1]

(ii) reduction

[1]

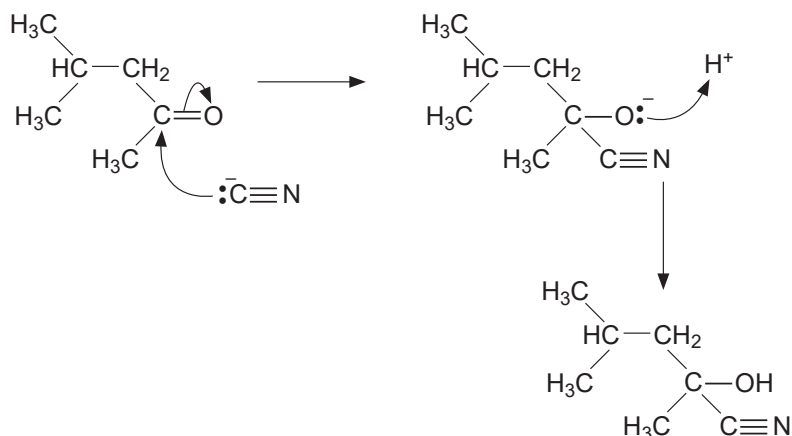
(iii) $\text{HOOCCH}_2\text{COOH} + 8[\text{H}] \rightarrow \text{HOCH}_2\text{CH}_2\text{CH}_2\text{OH} + 2\text{H}_2\text{O}$

[2]

AVAILABLE
MARKS

14

2 (a) (i)



3 curly arrows correct [1]

correct charges on CN^- , O^- , H^+ [1]

lone pairs on C of CN^- and O^- of intermediate [1]

[3]

(ii) nucleophilic addition

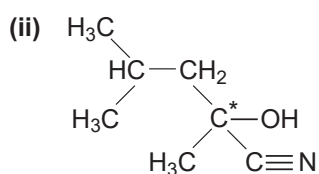
[1]

(iii) 4-methylpentan-2-one

[1]

(b) (i) an atom which has four different atoms or groups attached

[1]



[1]

(iii) CN^- attack equally probable from top or bottom/racemic mixture/equal concentrations of both optical isomers formed

[1]

(c) (i) 2-hydroxy-2,4-dimethylpentanoic acid

[2]

(ii) ammonia

[1]

(iii) ethanal

[1]

Section B

12

100

Total

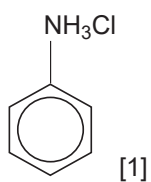
110

(b) (i) nitrobenzene [1]

(ii) tin [1]
concentrated hydrochloric acid [1] [2]

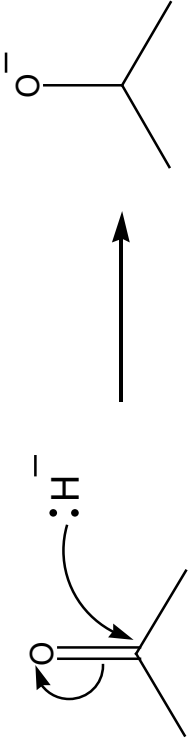
(iii) reduction [1]

(iv)

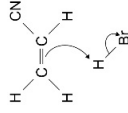
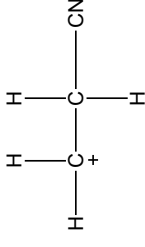
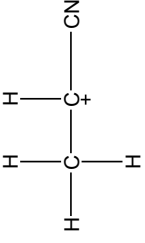


name: phenylammonium chloride [1] [2]

Question	Answers	Additional Comments/Guidelines	Mark
4 .1		M1 Structures of reactant species M2 Arrow from lp on O to C M3 Arrow from C=O bond to O M4 3 curly arrows and lp on intermediate Allow full marks for candidates who draw a second intermediate formed after formation of C=O and loss of Cl ⁻ then loss of H ⁺ Ignore any attempt to show the final products	4 (4 x AO1)
Question	Answers	Additional Comments/Guidelines	Mark
4 .2		M1 Structures of reactant species including + on N of ⁺ NO ₂ M2 Arrow from ring (inside hexagon) to N or + on N M3 Arrow from C-H bond into hexagon Apply list principle	4 (4 x AO1)

Question	Answers	Additional Comments/Guidelines	Mark
4 .3	 M4 Nucleophilic addition	M1 Arrow from lp on hydride to C M2 Arrow from C=O to O M3 Intermediate structure / Allow displayed or abbreviated structures Ignore any attempt to show further steps if correct Penalise further incorrect steps Apply list principle	4 (4 x AO1)

Question	Answers	Additional Comments/Guidelines	Mark
5 .1	3-bromopropanenitrile	Allow 3-bromopropane-1-nitrile	1 (AO1)

Question	Answers	Additional Comments/Guidelines	Mark
5 .2	This question is marked using Levels of Response. Refer to the Mark Scheme Instructions for Examiners for guidance.	Indicative Chemistry content Stage 1 Types of Isomers formed 1a CH_3CHBrCN 1b Exists as two Optical isomers / enantiomers Stage 2 Mechanism 2a 2 curly arrows  2b Intermediate structure primary carbocation OR  2c Alternative Intermediate structure secondary carbocation OR 	6 (3 x AO1, 3 x AO3)
	Level 3 5–6 marks All stages are covered and each stage is generally correct and virtually complete. Answer is communicated coherently and shows a logical progression from Stage 1 to Stages 2 and 3.		
	Level 2 3–4 marks All stages are covered but stage(s) may be incomplete or may contain inaccuracies OR two stages are covered and are generally correct and virtually complete. Answer is communicated mainly coherently and shows a logical progression from Stage 1 to Stages 2 and 3.		
	Level 1 1–2 marks Two stages are covered but stage(s) may be incomplete or may contain inaccuracies OR only one stage is covered but is generally correct and virtually complete. Answer includes isolated statements but these are not presented in a logical order.		
	0 mark Insufficient correct chemistry to gain a mark.		

	Stage 3 Optical isomerism 3a 2-bromo isomer has chiral carbon / C with four different groups / non superimposable mirror images OR $\begin{array}{c} \text{CH}_3 \\ \\ \text{C} \cdots \cdots \text{CN} \\ \\ \text{H} \end{array}$ $\begin{array}{c} \text{CH}_3 \\ \\ \text{C} \cdots \cdots \text{NC} \\ \quad \diagup \\ \text{H} \quad \text{Br} \end{array}$ 3b Optical because (secondary) C⁺ planar 3c So can be attacked from above or below	
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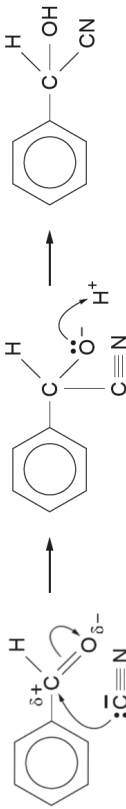
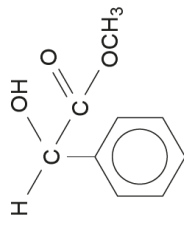
Question	Answers	Additional Comments/Guidelines	Mark
5 1.3	M1 KCN or NaCN M2 Aqueous AND ethanol (alcohol)	Penalise acid in M1	2 (2 x AO1)
Question	Answers	Additional Comments/Guidelines	Mark
5 1.4	M1 H ₂ and Ni/Pt/Pd M2 NCCH ₂ CH ₂ CN + 4H ₂ → H ₂ N(CH ₂) ₄ NH ₂	Allow LiAlH ₄ and (Dry) ether BUT <u>not</u> NaBH ₄ (ignore heat and pressure) Allow with 8[H]	2 (1 x AO1, 1 x AO2)

Question	Answers	Additional Comments/Guidelines	Mark
06.1	Use H_2SO_4	Allow HCl / H_3PO_4 Ignore conc / dilute	1 (AO1)
Question	Answers	Additional Comments/Guidelines	Mark
06.2	M1 Cool test 2 warm (water bath) M2 Gas is tested with lighted splint in test 3 Bubble into limewater	Allow heat / hot Allow no test on gas needed	2 (2 x AO3)
Question	Answers	Additional Comments/Guidelines	Mark
06.3	M1 J and M M2 Test 1 (Orange solution goes) green M3 M M4 Test 2 (Blue solution gives a brick) red precipitate M5 J and L M6 Test 3 (Colourless gas that turns) limewater cloudy M7 K M8 Test 4 (Orange solution goes) colourless	Allow (Brown-red/orange/red) Allow M6 Test 3 fizz / effervescence Allow (Brown/Brown-red/yellow/yellow-orange) Allow decolorises bromine	8 (8 x AO3)

Question	Answers	Additional Comments/Guidelines	Mark
7 1	$C_nH_{2n-2}O$	Allow $C_nH_{2n}CO$ or $(CH_2)_nCO$ or $C_nH_{2(n-1)}O$	1 (AO2)
Question	Answers	Additional Comments/Guidelines	Mark
7 2	M1 curly arrow from bond to O M2 lone pair and curly arrow from lp to H M3 curly arrow from C-H bond to C-C bond Step 1 Step 2	Allow other C-O bond breaking for M1	3 (3 x AO2)

Question	Marking details	Marks available				
		AO1	AO2	AO3	Total	Prac

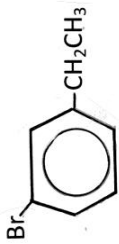
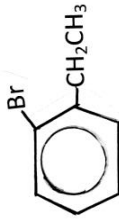
8	(c)	(i)	LiAlH_4	1			1		
		(ii)		1			1		
		(iii)	dehydration accept elimination			1	1		
	(d)		add NaHCO_3 / Na_2CO_3 (1) effervescence with the most acidic (1)		2		2		2
			Question 7 total	5	6	4	15	1	5

Question		Marking details	Marks available										
			AO1	AO2	AO3	Total	Maths	Prac					
9	(a)	I	 award (1) each for up to three of following curly arrows and addition of CN^- capture of H^+ partial and full charges dissociation of HCN into H^+ and CN^-						1	3		3	
	II	nucleophilic addition				1							
	(iii)	it is a racemic mixture / an equimolar mixture of the two enantiomers			1	1							
	(iv)	it does not contain a chiral centre / no asymmetric carbon atom		1		1							
	(v)	I it acts as a carboxylic acid 			1		2						
	II	it acts as an alcohol (giving an ester)		1		1							

Question		Marking details	Marks available					
			AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)	1			1		1
		orange to green (accept colourless to green)						
		(ii)						
		Indicative content • distillate placed in a separating funnel • add water to the distillate • add ethoxyethane to the funnel • equal volume of ethoxyethane as distillate (stopper) and shake • allow to separate • run off lower aqueous layer • treat ethoxyethane layer with anhydrous magnesium sulfate • remove magnesium sulfate (by filtration/decanting) • remove ethoxyethane by distillation • use of heating mantle		3	3	6	6	
		5-6 marks All the essential steps described in logical order to ensure a dry sample of pentan-2-one <i>The candidate constructs a relevant, coherent and logically structured account including all key elements of the indicative content. A sustained and substantiated line of reasoning is evident and scientific conventions and vocabulary are used accurately throughout.</i> 3-4 marks Most of the correct steps described (although order may not be correct); safe method involving no naked flames <i>The candidate constructs a coherent account including many of the key elements of the indicative content. Some reasoning is evident in the linking of key points and use of scientific conventions and vocabulary is generally sound.</i> 1-2 marks Some correct steps in separation method <i>The candidate attempts to link at least two relevant points from the indicative material. Coherence is limited by omission and/or inclusion of irrelevant materials. There is some evidence of appropriate use of scientific conventions and vocabulary.</i> 0 marks <i>The candidate does not make any attempt or give an answer worthy of credit.</i>						

Question		Marking details	Marks available					
			AO1	AO2	AO3	Total	Maths	Prac
	(iii)	no peak seen at 3200-3500 cm ⁻¹ (due to the OH group in the alcohol) accept no peak seen at 1000-1300 cm ⁻¹ (due to the C—O group in the alcohol)		1		1		
	(iv)	alkaline iodine / KI and NaOCl (1) yellow precipitate / solid forms (1) do not accept 2,4-DNPH test		2		2		2
		Question 10 total	3	12	5	20	0	9

Question		Marking details	Marks available					
			AO1	AO2	AO3	Total	Maths	Prac
10	(a)	(i)	(deep) blue Fehling's solution (1) gives a red / brown precipitate (1)	2		2		2
		(ii)	award (1) for name/formula of any aliphatic aldehyde e.g. propanal / CH ₃ CH ₂ CHO do not accept benzaldehyde / C ₆ H ₅ CHO	1		1		
	(b)	(i)	nucleophilic addition	1		1		
		(ii)	decomposition by water / dilute acid	1		1		
	(c)		award (1) each for any two of following loss of (volatile) ethanal incomplete oxidation / further oxidation side reactions		2	2		2
	(d)		moles of NaOH = $0.200 \times \frac{40.0}{1000} = 0.0080 \text{ mol}$ (1) moles of lactic acid reacted = 0.0080 mol mass of lactic acid reacted = $0.0080 \times 90.06 = 0.72 \text{ g}$ (1) volume of lactic acid = $\frac{0.72}{1.2} = 0.60 \text{ cm}^3$ percentage = $\frac{0.60}{75} \times 100 = 0.8\%$ (1)		2	3	1	

Question		Marking details	Marks available					
			AO1	AO2	AO3	Total	Maths	Prac
12	(a)			1		1		1
	(i)	bromine and iron(III) bromide / Br ₂ and FeBr ₃						
	(ii)	17	1			1		1
	(iii)	I award (1) for both of following structures   substitution can occur elsewhere in the ring (1)	1			2		
		II		1		1		
	(b)	alcoholic sodium hydroxide solution (1) a small molecule / HBr is removed (1)	1			2		2
	(c)	add aqueous bromine dropwise (1) bromine decolourised (1)	1			2		2

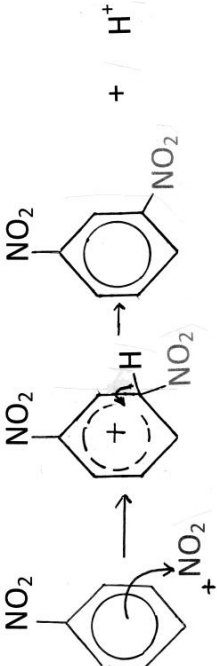
Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(d)	(i)	to react with benzoic acid (1) to give water soluble sodium benzoate (1) (other possible correct answers to be discussed at the examiners' conference)			2	2		
		(ii)	the density of trichloromethane is greater than the density of the aqueous layer		1		1		1
		(iii)	moles of phenylethene = $\frac{3.00}{104} = 0.289 \text{ mol}$ (1) moles of phenyloxirane = $\frac{2.50}{120} = 0.208 \text{ mol}$ (1) mole ratio is 1:1 percentage yield = $\frac{0.208}{0.289} \times 100 = 71.9\%$ (1)					2	
	(e)		a peak would be seen between 1650 and 1750 cm^{-1} (1) indicating the presence of the C=O group (1)	2			2		

Question		Marking details		Marks available															
				AO1	AO2	AO3	Total	Maths	Prac										
	(f)	<table><tr><td>Test</td><td>phenylethanal $\text{C}_6\text{H}_5\text{CH}_2\text{CHO}$</td><td>phenylethanone $\text{C}_6\text{H}_5\text{C(O)CH}_3$</td></tr><tr><td>alkaline iodine</td><td>no observable reaction</td><td>yellow precipitate formed</td></tr><tr><td>Tollens' reagent</td><td>silver mirror formed</td><td>no observable reaction</td></tr><tr><td>warming with acidified potassium dichromate</td><td>solution turns from orange to green</td><td>no observable reaction</td></tr></table> award (3) for all correct award (2) for any four or five correct award (1) for any two or three correct						Test	phenylethanal $\text{C}_6\text{H}_5\text{CH}_2\text{CHO}$	phenylethanone $\text{C}_6\text{H}_5\text{C(O)CH}_3$	alkaline iodine	no observable reaction	yellow precipitate formed	Tollens' reagent	silver mirror formed	no observable reaction	warming with acidified potassium dichromate	solution turns from orange to green	no observable reaction
Test	phenylethanal $\text{C}_6\text{H}_5\text{CH}_2\text{CHO}$							phenylethanone $\text{C}_6\text{H}_5\text{C(O)CH}_3$											
alkaline iodine	no observable reaction							yellow precipitate formed											
Tollens' reagent	silver mirror formed							no observable reaction											
warming with acidified potassium dichromate	solution turns from orange to green							no observable reaction											
		6	12	2	20	2	10												
		Question 8 total																	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
14	(b)	I	$\text{ClCH}_2(\text{CH}_2)_2\text{CH}_2\text{Cl} + 2\text{KCN} \rightarrow \text{NC}(\text{CH}_2)_4\text{CN} + 2\text{KCl}$		1		1		
		II	CN^-	1			1		
		III	award (1) for any of following lithium tetrahydroaluminate(III) / LiAlH_4 hydrogen with Ni catalyst sodium and ethanol	1			1		1

Question	Marking details				Marks available					
	AO1	AO2	AO3	Total	Maths	Prac				
15	(v)	I	award (1) for appropriate charges award (1) for curly arrows compound W				2	2	2	
	II	initial attack is by the nucleophile CN^- (1) HCN adds across the $\text{C}=\text{O}$ bond (1)				2		2		

Question		Marking details	Marks available					
			AO1	AO2	AO3	Total	Maths	Prac
(b)	(i)	lithium tetrahydridoaluminate(III) / LiAlH ₄ (1) hydrogen is added / oxygen is lost (1)	2			2		
	(ii)	benzene-1,4-dioic acid accept terephthalic acid		1		1		
(c)	(i)	$ \begin{array}{c} \text{COO}^-\text{Na}^+ \\ \\ \text{CH}_2 \\ \\ \text{COO}^-\text{Na}^+ \end{array} + 2\text{NaOH} \longrightarrow 2\text{Na}_2\text{CO}_3 + \text{CH}_4 $		2		2		
	(ii)	sodium hydrogencarbonate / NaHCO ₃ (1) accept sodium carbonate / Na ₂ CO ₃ reaction complete when no more effervescence seen (1) phenol does not react with sodium hydrogencarbonate (to give carbon dioxide) (1)		3		3		3
	(iii)	no purple coloration seen			1	1		1
		Question 11 total	5	12	1	18	1	4

Question		Marking details	Marks available					
(b)	(i)		AO1	AO2	AO3	Total	Maths	Prac
16				2		2		
	(ii)	I	1			1		1
		Fe / Sn and hydrochloric acid						
		II	1			1		1
		NaOH / sodium hydroxide						
	(iii)	HNO ₂ / NaNO ₂ , HCl / nitric(III) acid / sodium nitrate(III) and hydrochloric acid 0-10°C	2			2		2
	(iv)	amine group in the azo dye has a lone pair of electrons on the nitrogen atom and acts as an electron pair donor / proton acceptor			1	1		

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